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sravya@Sravya1955:~/os$ cat filecopy1.c
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>
#include <sys/time.h>

#define BUFFER_SIZE 4096

double get_time() {
    struct timeval tv;
    gettimeofday(&tv, NULL);
    return tv.tv_sec + tv.tv_usec / 1000000.0;
}

/* Low-level system call copy */
void copy_low_level(const char *src, const char *dest) {
    int fd_src, fd_dest;
    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    fd_src = open(src, O_RDONLY);

    if (fd_src < 0) {
        perror("open source");
        exit(1);
    }

    fd_dest = open(dest, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd_dest < 0) {
        perror("open destination");
        close(fd_src);
        exit(1);
    }

    while ((bytes_read = read(fd_src, buffer, BUFFER_SIZE)) > 0) {
        bytes_written = write(fd_dest, buffer, bytes_read);

        if (bytes_written != bytes_read) {
            perror("write");
        }
    }
}
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        close(fd_src);
        close(fd_dest);
        exit(1);
    }
}

close(fd_src);
close(fd_dest);
}

/* Standard library copy */
void copy_standard(const char *src, const char *dest) {
    FILE *source;
    FILE *destination;
    char buffer[BUFFER_SIZE];
    size_t bytes_read;

    source = fopen(src, "rb");

    if (source == NULL) {
        perror("fopen source");
        exit(1);
    }

    destination = fopen(dest, "wb");

    if (destination == NULL) {
        perror("fopen destination");
        fclose(source);
        exit(1);
    }

    while ((bytes_read = fread(buffer, 1, BUFFER_SIZE, source)) > 0) {
        fwrite(buffer, 1, bytes_read, destination);
    }

    fclose(source);
    fclose(destination);
}

int main() {

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const char *source = "input.txt";
const char *low_level_dest = "copy_low.txt";
const char *standard_dest = "copy_standard.txt";

double start, end;

printf("=== Low-Level File Copy ===\n");

start = get_time();
copy_low_level(source, low_level_dest);
end = get_time();

printf("Low-level copy completed.\n");
printf("Time: %.6f seconds\n\n", end - start);

printf("=== Standard Library File Copy ===\n");

start = get_time();
copy_standard(source, standard_dest);
end = get_time();

printf("Standard library copy completed.\n");
printf("Time: %.6f seconds\n", end - start);

return 0;
}

```

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LinuxLab +ito_project
sravya@Sravya1955:~/os$ ./filecopy1
=== Low-Level File Copy ===
Low-level copy completed.
Time: 0.001405 seconds

=== Standard Library File Copy ===
Standard library copy completed.
Time: 0.000426 seconds

```

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GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <fcntl.h>
#include <unistd.h>

int main() {
    int fd;

    fd = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd < 0) {
        perror("open");
        return 1;
    }

    if (dup2(fd, STDOUT_FILENO) < 0) {
        perror("dup2");
        close(fd);
        return 1;
    }

    close(fd);

    printf("Hello from redirected output!\n");
    printf("This output is written into the file.\n");

    return 0;
}

```

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sravya@Sravya1955:~/os$ gcc redirect.c -o redirect
sravya@Sravya1955:~/os$ ./redirect
sravya@Sravya1955:~/os$ cat output.txt
Hello from redirected output!
This output is written into the file.
sravya@Sravya1955:~/os$ |

```